

Aloysius Allen Darmawan

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EDUCATION

Monash University

Feb. 2022 – Nov. 2026 (Expected)

Bachelor of Science (Pure Mathematics, Physics)/Bachelor of Computer Science

Melbourne, VIC

- Weighted Average Mark (WAM): Distinction average

EXPERIENCE

Machine Learning Engineer Intern

Jun. 2025 - Jan. 2026

Geotesta

Melbourne, VIC

- Developed a PyTorch image classifier for tunnel face rock structure recognition, integrating Google Vision and REST APIs via FastAPI, achieving 90% validation accuracy and reducing manual review time by 30%
- Deployed a full-stack website application for tailings dam safety analysis with an integrated AI chatbot, built using Flask and Bootstrap, streamlining workflows for 20+ engineers across multiple firms

Backend Engineer Intern

Nov. 2024 – Feb. 2025

ANZ Plus

Melbourne, VIC

- Streamlined CI/CD pipelines by building GitHub Actions workflows for release tagging and deployment validation across 5+ environments, reducing deployment errors by 40%
- Developed and deployed Go cloud-native microservices for real-time credit decisioning, automating SQL generation to query BigQuery datasets with 100M+ records resulting in scalable, data-driven lending decisions

Student Software Engineer

Aug. 2023 – Feb. 2025

Monash Connected Autonomous Vehicle

Melbourne, VIC

- Implemented ROS 2 launch systems in Python and C++ to synchronise 8 LiDAR sensors on a Nissan e-NV200 van, achieving deterministic and real-time sensor fusion for robust autonomy of the van
- Developed an autonomous valet parking system using the Autoware (ROS 2) stack for sub-50ms control latency

COMMUNITY AND LEADERSHIP

Mental Health Immersive Technologies Developer

Feb. 2026 – Apr. 2026

EUGLOH

Paris, France

- While on semester exchange at Lund University, selected for an international program at University of Paris-Saclay to co-develop a VR/AR environment bridging art therapy and mental health using Unity, TouchDesigner, and C#

University Open Day Student Ambassador

Aug. 2024 – Aug. 2025

Monash University

Melbourne, VIC

- Represented the Faculty of Information Technology as a student ambassador at outreach events such as Open Day, guiding 300+ prospective students and their families through degree offerings, job prospects and university life

Mathematics & Physics Teaching Assistant

Aug. 2023 – Oct. 2023

Embrace Education

Melbourne, VIC

- Designed and facilitated workshops covering high school mathematics and physics subjects for 50+ students from low socio-economic backgrounds; received consistent positive feedback for engaging teaching

PROJECTS

CME Sentinel | *Planned tech stack: Linux, C/C++, Java, Python, PostgreSQL, AWS, Bash*

GitHub

- Building 24/7 space weather platform tracking NASA coronal mass ejections that predict geomagnetic storms; aiming to deliver 99.7% uptime via automated monitoring enabling real-time space weather forecasting

Avian Annotator | *React, Spring Boot, Node.js, PostgreSQL, AWS S3, Docker*

GitHub

- Oversaw the development of an accessible web-based bird image annotation platform allowing ornithology researchers and citizen scientists to collaboratively create high-quality training datasets for AI models

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, C, C++, C#, Go, SQL, HTML, CSS, MATLAB

Frameworks/Libraries: React.js, Node.js, Spring Boot, FastAPI, Flask, PyTorch, Bootstrap, ROS 2

Tools/Platforms: Git, Docker, PostgreSQL, Google Cloud Platform, Jira, Visual Studio Code

Mathematics/Physics: Linear Algebra, Probability, Statistics, Graph Theory, Calculus, Fluid Mechanics